EXPERIMENT 5

Aim:

1. Write the programs to display the following morphological transformation of the input grayscale image:
   a. Erosion
   b. Dilation
   c. Opening
   d. Closing
   e. Morphological Gradient
2. Prepare a program for detecting the edges of an image using the Laplacian operator:

$$K = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

3. Write a program to perform the process of Canny Edge Detection for a grayscale image as input.
4. Prepare a program for displaying all the types of image thresholding process:
   a. cv2.THRESH_BINARY
   b. cv2.THRESH_BINARY_INV
   c. cv2.THRESH_TRUNC
   d. cv2.THRESH_TOZERO
   e. cv2.THRESH_TOZERO_INV

Code:

```
import cv2
import numpy as np

path = '/Users/vishrutgrover/coding/sem6/cv/lab5/'
img = cv2.imread(path + 'gojo.webp')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
_, binary = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)
k = np.ones((5, 5), np.uint8)

# 1. Morphological transformations
cv2.imwrite(path + '1a_erosion.png', cv2.erode(binary, k))
cv2.imwrite(path + '1b_dilation.png', cv2.dilate(binary, k))
```

```
cv2.imwrite(path + '1c_opening.png', cv2.morphologyEx(binary, cv2.MORPH_OPEN, k))
cv2.imwrite(path + '1d_closing.png', cv2.morphologyEx(binary, cv2.MORPH_CLOSE, k))
cv2.imwrite(path + '1e_gradient.png', cv2.morphologyEx(binary, cv2.MORPH_GRADIENT, k))

# 2. Laplacian edge detection
cv2.imwrite(path + '2_laplacian.png', cv2.Laplacian(gray, cv2.CV_64F, ksize=1))

# 3. Canny edge detection
cv2.imwrite(path + '3_canny.png', cv2.Canny(gray, 100, 200))

# 4. Image thresholding
for name, thresh_type in [('binary', cv2.THRESH_BINARY), ('binary_inv',
cv2.THRESH_BINARY_INV),
                          ('trunc', cv2.THRESH_TRUNC), ('tozero', cv2.THRESH_TOZERO),
                          ('tozero_inv', cv2.THRESH_TOZERO_INV)]:
    _, out = cv2.threshold(gray, 127, 255, thresh_type)
    cv2.imwrite(path + f'4_{name}.png', out)
```

Output:

Original Image

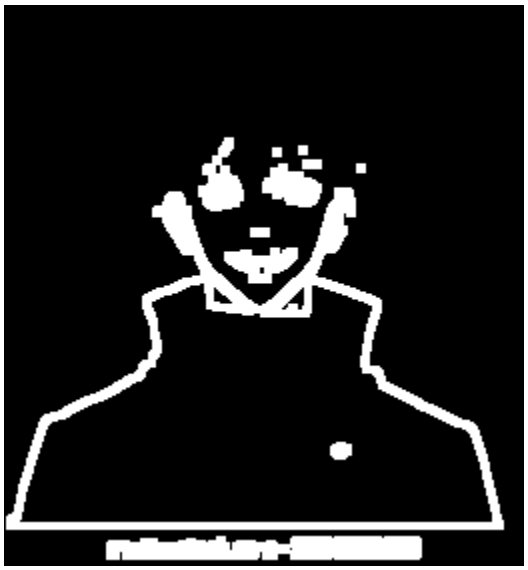
1.
    a. Erosion

b. Dilation

c. Opening

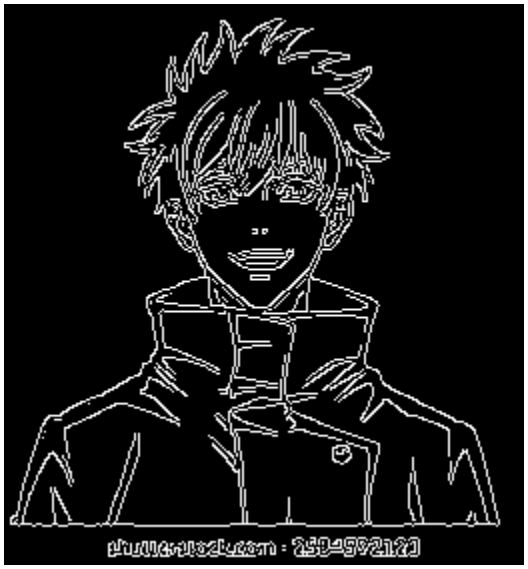
d. Closing

e. Morphological Gradient



2. Edges of an image using the Laplacian operator:

3. Canny Edge Detection for a grayscale image as input.



4. Prepare a program for displaying all the types of image thresholding process:

a. cv2.THRESH_BINARY

b. cv2.THRESH_BINARY_INV

c. cv2.THRESH_TRUNC

d. cv2.THRESH_TOZERO

e. cv2.THRESH_TOZERO_INV